

# SIMATS ENGINEERING

## ASSESSMENT TOOL 3

**COURSE NAME: INTERNET PROGRAMMING COURSE**  
**CODE: CSA4305**

### COURSE OUTCOME COVERED

**CO 4:** Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework. (L4)

Assessment Tool: Class Test

Weightage: 10%

### QUESTIONS:

**Total Marks: 20**

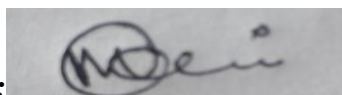
- Explain the fundamental concepts used in web development by defining the following terms with suitable explanations: **XML, JSP, MVC architecture, XSLT, XML Schema, JSTL, PHP, Database Connectivity, and Parser**. Your answer should clearly describe the purpose and role of each concept in web applications.
- Explain how the above technologies are interconnected in building a web application. Describe how XML is structured and validated, how JSP and PHP are used for server-side processing, how MVC architecture organizes the application, and how database connectivity and parsers contribute to data processing and communication.

### ***Code of Conduct:***

*I P Madhavi/192372133 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student:**



**RUBRICS FOR CALCULATION:**

| <b>Criteria</b>                | <b>Weightage</b> | <b>Level 4 – Exemplary</b>                                     | <b>Level 3 – Proficient</b>             | <b>Level 2 – Developing</b>          | <b>Level 1 – Beginning</b>              |
|--------------------------------|------------------|----------------------------------------------------------------|-----------------------------------------|--------------------------------------|-----------------------------------------|
| <b>Concept Understanding</b>   | 25%              | Clearly explains all concepts with complete accuracy and depth | Minor gaps in understanding             | Basic understanding with some errors | Incorrect or very limited understanding |
| <b>Content Coverage</b>        | 25%              | Covers all required topics comprehensively                     | Covers most topics with minor omissions | Covers some topics only              | Very few topics covered                 |
| <b>Technical Accuracy</b>      | 20%              | All explanations are technically correct and precise           | Minor technical errors                  | Some incorrect or vague explanations | Major technical errors                  |
| <b>Integration of Concepts</b> | 15%              | Effectively connects and relates concepts logically            | Some connections made with minor gaps   | Limited connections between concepts | No clear relationship between concepts  |

## ① Fundamental concepts used in web development

web development involves technologies that help create, manage, display and process information in web applications. the following are the important concepts

### XML (Extensible markup language):

XML is a markup language used to store and transport structured data in a human-readable and machine readable format.

#### Purpose in web application

- \* Stores data independently from presentation.
- \* Enables data Exchange between different systems
- \* used in APIs, configuration files, and web services

#### Example XML Structure:

```
<student>
```

```
  <id>101</id>
```

```
  <name>madhavi</name>
```

```
  <department>CSE-A1</department>
```

```
</student>
```



## Advantages:

- \* platform Independent.
- \* Easy to read and write.
- \* supports hierarchical data structure.

## 2. JSP (Java Server Pages)

\* JSP is a Server-side technology used to create dynamic web pages using Java.

### Purposes:

- \* Generates HTML dynamically.
- \* processes user requests.
- \* connects with Java classes and databases.

### Features:

- \* Java codes embedded inside HTML.
- \* Execute on web server.
- \* produces dynamic content.

3. MVC Architecture

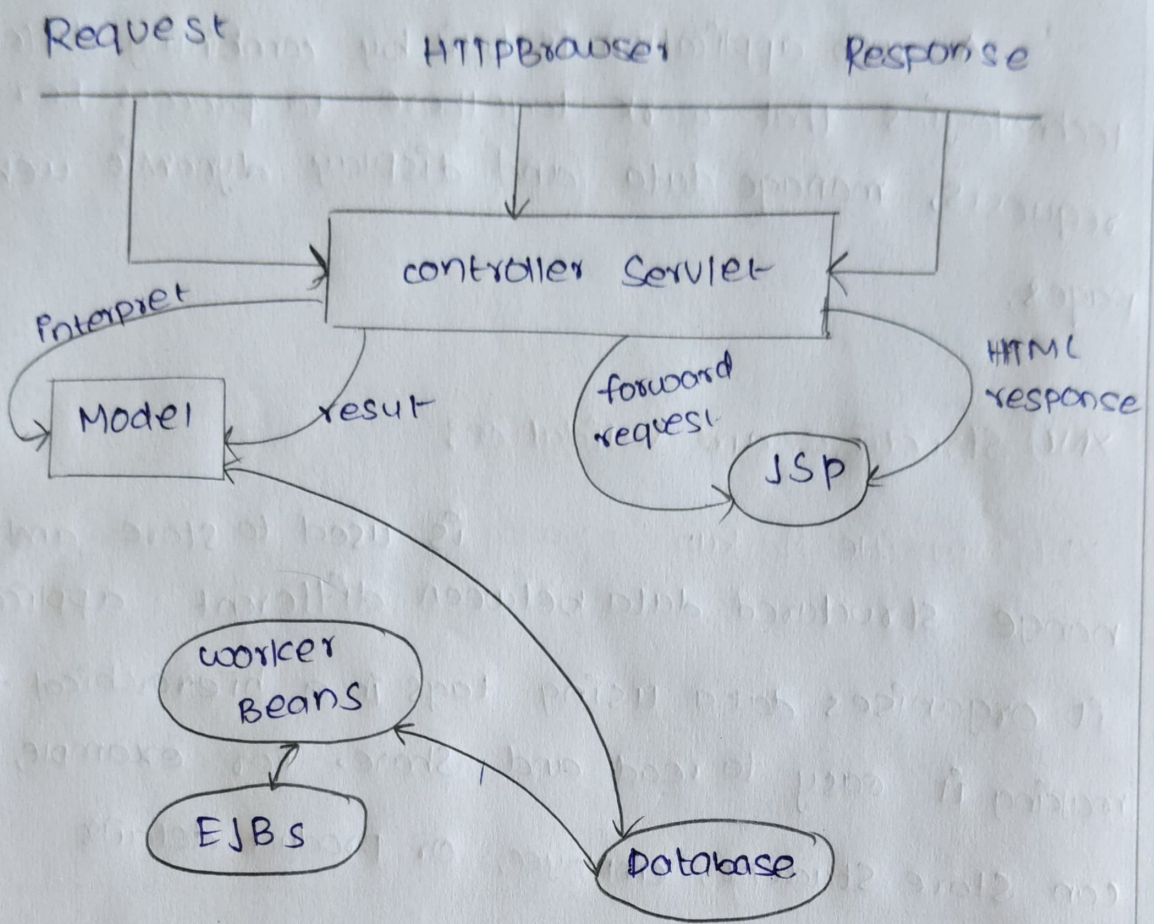
4. XSLT (Extensible stylesheet language transformation)

5. XML Schema.

6. JSTL

7. PHP





### 8. Database connectivity:

Database connectivity is the process of connecting a web applications to a databases. It allows storing, retrieving, updating, and deleting data using SQL.

### 9. parser:

A parser is a program that reads and processes to XML or HTML documents. It extracts and validates Structured documents.



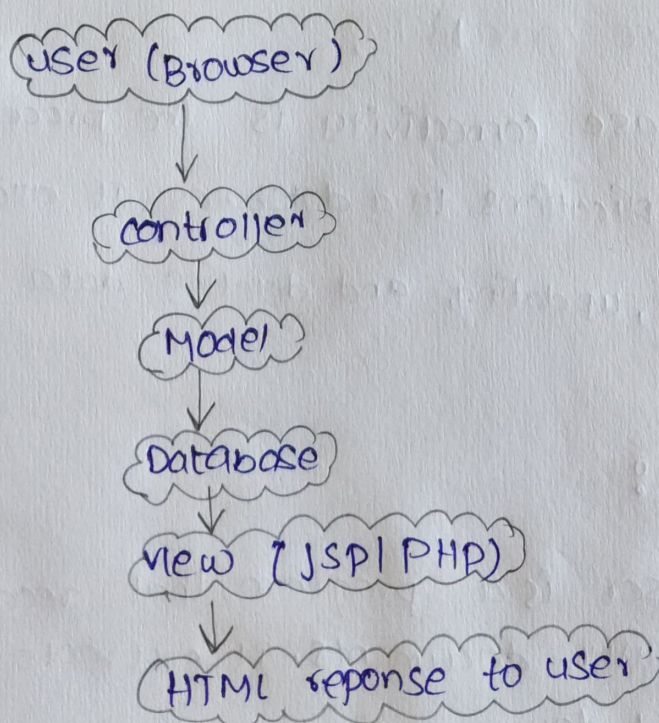
② Explain how the above technologies are interconnected in building a web application.

A web application is built by combining different technologies that work together to process requests, manage data and display dynamic web pages.

### XML Structure and Validation

XML (Extensible Markup language) is used to store and exchange structured data between different applications. It organises data using tags in a hierarchical format, making it easy to read and share. For example, XML can store student, employee, or product details.

Overall web applications flow (MVC + Database)





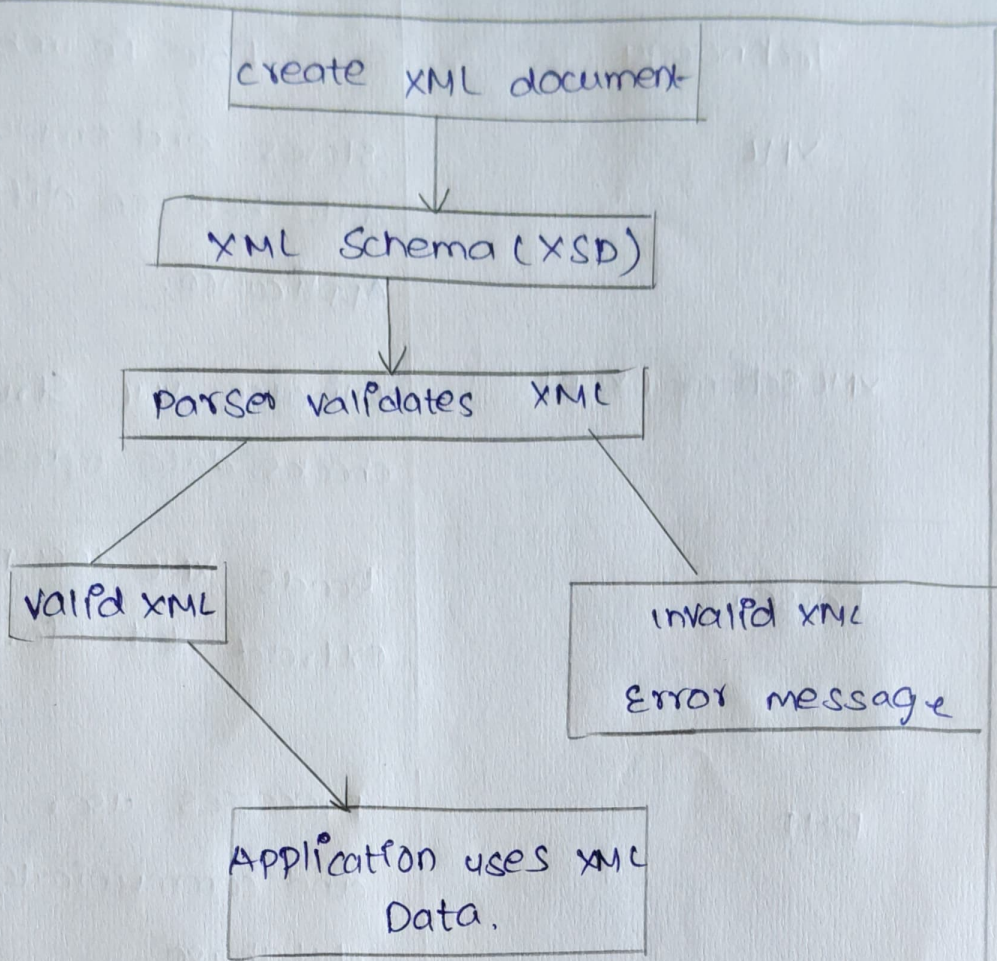
| Technology       | Roles in web Application.                                          |
|------------------|--------------------------------------------------------------------|
| XML              | stores and exchange structured data between different Application. |
| XML Schema (XSD) | validates XML Structure and checks data types and roles.           |
| Parser           | Reads XML documents and extract data for processing                |
| PHP              | processes user requests and communicates with the database         |

## MVC Architecture

The MVC Model (Model-view-controller) architecture organises a web application into three separate Model.

- \* Model : Manages the application's data and logic
- \* View : Displays information to the user through pages.
- \* Controller : Receives user requests, processes them, and co-ordinates communication b/w the Model and View.





## XML validation:

| Steps            | Description.                        |
|------------------|-------------------------------------|
| XML document     | contains structured data using tags |
| XML Schema (XSD) | define rules and validates XML.     |
| parser           | Reads and checks XML Structure      |
| Output           | XML is accepted.                    |